# Age-Normative EEG Classification of Mild Cognitive Impairment and Dementia Using State-Aware Learning

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Abstract—Resting-state EEG is an inexpensive candidate for screening mild cognitive impairment (MCI) and dementia, but accuracies reported on public benchmarks are hard to interpret. We revisit the three-class CAUEEG-Dementia benchmark (Normal / MCI / Dementia, 1,187 recordings) with an emphasis on what the data allow a model to learn. First, CAUEEG event annotations count paused recording time that is absent from the EDF signal; a simple correction reduces the end-of-recording misalignment in the 293 paused recordings from a median of 33 s (up to 678 s) to 3 s, enabling per-second labelling of eyes-closed, eyes-open and photic-stimulation states. Second, using the corrected clock we build SANF-Net, a compact (143k-parameter) network that samples EEG by behavioural state and fuses it with 48 age-normative clinical biomarkers and age through interpretable modality gates. Third, we identify a protocol confound: only 12% of community-recruited normal recordings include photic stimulation, versus 90–92% of every other group, and these volunteers are also 8–15 years younger. Under patient-disjoint, repeated five-fold cross-validation, SANF-Net reaches 59.5% accuracy and a macro AUC of 0.778, matching logistic regression on the same normative biomarkers (59.3%, 0.778) and exceeding EEGNet (52.9%, 0.707) and a CEEDNet-style 1D-ResNet (57.7%, 0.752) trained under the same protocol. Every biomarker-based model separates community normals from MCI well (AUC 0.84–0.95) but subjective memory impairment (SMI) from MCI poorly (0.62–0.66). After 1:1 age matching, discrimination of SMI from MCI falls to 0.54–0.57, and community normals with the same photic protocol as MCI are not distinguished from them at all (0.50–0.59). External testing without retraining gives an MCI-versus-normal AUC of 0.67 (0.70 with label-free site calibration) on the pooled Isfahan cohorts and 0.58 on GENEEG. Much of the apparent EEG signal for MCI on CAUEEG is age and recording protocol; we argue that confound-controlled evaluation against strong tabular baselines should be standard for EEG dementia benchmarks.

Index Terms—EEG, dementia, mild cognitive impairment, normative modelling, shortcut learning, deep learning, CAUEEG, external validation

## I. Introduction

Dementia affects about 57 million people worldwide [1], and mild cognitive impairment (MCI) is its main prodromal stage [2]. Neuropsychological testing, PET and CSF biomarkers are costly or invasive, whereas scalp EEG is cheap, portable and widely available. Alzheimer's-type dementia produces well-documented EEG changes: slowing of the background rhythm, reduced posterior alpha, increased theta and delta power, and reduced coherence [3]–[5].

The CAUEEG dataset [6] is one of the largest public clinical EEG collections with a Normal / MCI / Dementia benchmark and a deep-learning baseline (CEEDNet). Results on it are, however, difficult to interpret for four reasons. (i) Clinical EEG slowing is read during eyes-closed rest, yet recordings also contain eyes-open periods, photic stimulation and hyperventilation, and random crops mix these states. (ii) Per-segment scores, or splits that share patients, inflate accuracy relative to per-recording, patient-disjoint evaluation [7]. (iii) A 118-recording test set has a 95% confidence interval of roughly ±9 accuracy points [8]. (iv) Deep models are rarely compared against a well-tuned tabular model of established EEG markers under the same protocol, and rarely tested on another site.

CAUEEG includes event annotations that could address (i), but we found that their time indices include paused periods that the EDF signal omits, so naive use misplaces state boundaries by up to minutes. Correcting this makes the behavioural state of every second known, which enables state-aware modelling and also reveals that the cohorts were recorded with different protocols.

Contributions. (1) A verified correction of CAUEEG event timing and a per-second state and artifact timeline (Sec. IV-B). (2) SANF-Net, a state-aware, age-normative fusion network using reactivity and photic-driving biomarkers that are only computable with the corrected clock (Sec. IV-D). (3) The identification of a photic-stimulation protocol confound between the Normal subgroups, and an age-matched analysis that separates it from the age confound (Sec. V-C). (4) A patient-disjoint, per-recording evaluation with repeated five-fold cross-validation, bootstrap confidence intervals, tabular and deep baselines trained under the same protocol, ablations, and external testing on two independent MCI cohorts (Secs. V–V-E).

## II. Related Work

EEG markers of cognitive decline. Spectral slowing (theta/alpha and delta/alpha ratios), lower alpha peak

frequency, reduced alpha reactivity to eye opening, and loss of inter-regional coherence are consistently reported in Alzheimer's disease and, more weakly, in MCI [3]–[5].

Machine learning on CAUEEG. Kim et al. [6] released CAUEEG with CEEDNet, a 1D/2D-CNN and transformer family trained on random 10 s crops with age as an input, and reported 70.2% (single model) and 74.7% (ensemble) three-class test accuracy per recording. CEFA-Net [9] combines a three-scale CNN with a branch of clinical EEG ratios and a focal loss, reporting 81.0% for dementia staging with per-segment evaluation and 73.8% on a patient-disjoint test. A microstate-based SVM reports 89% accuracy on CAUEEG [10]. These numbers come from different evaluation units and splits and are not directly comparable to each other or to ours. Mishra et al. [11] contrasted CAUEEG with GENEEG, a general-practice cohort, and found high-variance, poorly calibrated models when training data are small, poor transfer between the two datasets, and strongly overlapping MCI and control feature distributions. Compact architectures such as EEGNet [12] are common baselines for small clinical datasets.

Normative modelling. Normative models describe each subject as a deviation from a reference population conditioned on covariates such as age [13]. This separates normal ageing from pathology, which matters for EEG because the background rhythm also slows with healthy ageing.

Shortcut learning and evaluation pitfalls. Classifiers frequently exploit acquisition artifacts instead of the target signal [14]; in medical imaging, differences in scanner or protocol between cohorts are a well-known source [15], [16]. Record-wise instead of subject-wise splits inflate performance [7].

## III. Data

### A. CAUEEG-Dementia

CAUEEG [6] contains clinical 19-channel 10–20 scalp EEG recorded at 200 Hz at Chung-Ang University Hospital, with event annotations (eyes open/closed, photic stimulation, hyperventilation, artifacts, pauses). We use the official CAUEEG-Dementia split: 950 training, 119 validation and 118 test recordings (459 Normal, 417 MCI, 311 Dementia in total). Using the dataset's symptom codes, we divide the Normal class into community-recruited normal volunteers, patients presenting with subjective memory impairment (SMI) whose testing was normal, and 13 Normal recordings with neither code ("other"), which remain in the Normal class but are left out of subgroup analyses. The subgroups differ markedly (Table I). We also report the official no-overlap test subset (90 recordings), which excludes patients with recordings in the training split.

TABLE I
CAUEEG-Dementia subgroups (all 1,187 recordings). Durations are clean minutes per recording after artifact masking.

| Group | $n$ | Age (years) | Photic (%) | EO (%) | Length (min) | EC (min) | Photic (min) |
|---|---|---|---|---|---|---|---|
| Normal: community | 260 | $61.7 \pm 8.3$ | 12 | 63 | 10.2 | 5.7 | 0.1 |
| Normal: SMI | 186 | $69.6 \pm 9.3$ | 90 | 84 | 13.8 | 5.6 | 0.7 |
| Normal: other | 13 | $68.4 \pm 7.2$ | 69 | 85 | 13.2 | 5.5 | 0.6 |
| MCI | 417 | $73.8 \pm 7.8$ | 92 | 87 | 13.8 | 5.8 | 0.8 |
| Dementia | 311 | $76.6 \pm 8.1$ | 92 | 80 | 14.0 | 6.6 | 0.8 |

Photic / EO: share of recordings with usable photic-stimulation / eyes-open data. Length: total recording.

### B. External cohorts

For external testing without retraining we use two public MCI datasets: Isfahan [17], [18], two releases of 19-channel, 256 Hz eyes-closed resting EEG (2016: 16 normal / 11 MCI; 2019: 16 normal / 18 MCI), and GENEEG [11], general-practice recordings (35 control / 28 MCI) acquired at 250 Hz during an eyes-closed P300 protocol, with T5/T6 unavailable (interpolated). Neither provides state annotations or per-subject ages, so all segments are treated as one state and a fixed age is imputed for every subject (68 and 73.4 years), which removes age as a source of information. Recordings are re-referenced to the common average as in CAUEEG. MCI versus normal is scored with $1 - P(\text{Normal})$.

## IV. Methods

### A. Pre-processing

Signals are band-pass filtered (0.5–45 Hz, 4th-order Butterworth, zero phase), clipped at $\pm 3000\,\mu$V and resampled to 100 Hz. The first 10 s are discarded for electrode settling.

### B. Event-clock correction and state timeline

Each CAUEEG event is a sample index at 200 Hz with a text label. We found that these indices count wall-clock time including periods in which the recording was paused, whereas the EDF signal contains only the recorded portions joined end-to-end. Event $i$ therefore maps to signal position

$$\hat{t}_i = t_i - \sum_{p:\, r_p \le t_i} (r_p - s_p), \tag{1}$$

where $s_p$ and $r_p$ are the indices of the Paused and Recording resumed events of pause $p$; events annotated during a pause are placed at the join. Every recording ends with a Paused event, which should coincide with the end of the signal and so gives a check on alignment. Of the 1,187 recordings, 293 contain at least one pause. In these, the raw final event lies a median of 33 s (90th percentile 90 s, maximum 678 s) from the signal end; after correction the median is 3 s and the 90th percentile 6 s (Fig. 1). Across all recordings, 1,184 of 1,186 align within 6 s after correction; the single large residual (87 s) is a recording with six pauses.

From the corrected events we build a per-second timeline with five states (eyes-closed, eyes-open, photic, hyperventilation, unknown). Photic and hyperventilation

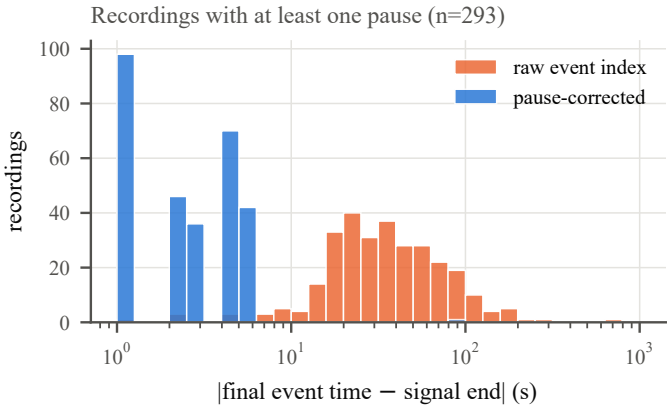



Fig. 1. Offset between the final Paused event and the end of the EDF signal for recordings containing pauses, before (orange) and after (blue) subtracting paused time. Offsets below 1 s are plotted at 1 s.

override the eye state, and the 60 s after hyperventilation remain labelled hyperventilation. A per-second clean mask removes 1 s before and 3 s after every state change (absorbing the residual offset), 3 s around pause joins (filter ringing at the discontinuity), 2 s before to 3 s after point artifacts (movement, coughing, swallowing, talking, etc.), and 30 s after drowsiness, sleep or snoring annotations. Hyperventilation is labelled only so that it can be excluded: it is present in few recordings (0.01–0.4 clean minutes on average per group) and induces EEG slowing in healthy subjects that would mimic pathology, so it is used neither for segments nor for biomarkers.

### C. Age-normative clinical biomarkers

For each recording we compute 48 biomarkers from Welch spectra of 4 s windows (2 s step, at most 150 windows per state):

- Eyes-closed (39): log relative power in $\delta$ (1–4 Hz), $\theta$ (4–8), $\alpha$ (8–13) and $\beta$ (13–30) for five regions; theta/alpha, delta/alpha and slow/fast ratios; fronto-posterior alpha ratio; alpha and beta hemispheric asymmetry; posterior alpha centre of gravity and peak frequency; spectral entropy; median frequency; total power; Hjorth mobility and complexity; inter-hemispheric and fronto-posterior coherence in $\theta$, $\alpha$ and $\beta$.
- Eyes-open (5): global relative band powers and posterior alpha centre of gravity.
- Reactivity and driving (2): posterior alpha reactivity, $\log(P_\alpha^{EC}/P_\alpha^{EO})$; and occipital photic-driving SNR, the log ratio of O1/O2 power within ±0.5 Hz of the flash frequency to power 1–3 Hz away, averaged over stimulation blocks up to 25 Hz.
- Availability (2): indicators has_eo and has_photic.

The set includes the eight clinical ratios used by CEFA-Net [9]. The reactivity and photic-driving markers can only be computed with a correct event clock. Recording length is not an input; each recording contributes a fixed number of segments.

Because EEG slows with healthy ageing, a raw biomarker confounds age with impairment. For every biomarker $x_j$ we fit a linear age model on the training Normal recordings only and replace $x_j$ by its robust deviation

$$z_j = \frac{x_j - f_j(\text{age})}{1.4826\,\text{MAD}_j}, \tag{2}$$

where $\text{MAD}_j$ is the median absolute deviation of the normal residuals. Values are clipped to ±6; a missing state gives $z_j = 0$ (typical for age), and the indicator records the absence. In cross-validation the normative model is refitted within each fold.

### D. SANF-Net

SANF-Net (Fig. 2) has three input branches.

EEG branch. A recording is a bag of 10 s segments (19 × 1000 samples) drawn from clean seconds by state: 60% eyes-closed, 25% eyes-open and 15% photic (12 segments per recording in training, 24 at evaluation). A shared encoder applies three parallel temporal convolutions (0.11, 0.25 and 0.51 s kernels, matching beta, alpha and delta–theta time scales), a depthwise spatial convolution over the 19 channels, a separable convolution with squeeze-and-excitation (SE), a temporal convolution with a second SE block, and mean and standard-deviation pooling to a 128-d embedding $h$. Each embedding is modulated by its state $s$ with feature-wise linear modulation (FiLM) [19], $h' = h \odot (1 + \gamma_s) + \beta_s$, initialised to identity, and gated attention pooling [20] combines the segments of a recording.

Biomarker and age branches. The 48 normative $z$-scores pass through a two-layer MLP (64-d) and standardised age through a small MLP (16-d). Age is an explicit input so that its contribution can be measured and ablated.

Modality-gated fusion. Each branch is projected to 128-d; a linear layer over the concatenated projections yields softmax gates $g \in \Delta^3$, and the fused vector is $\sum_m g_m h_m$. The gates show how much each prediction relied on EEG, biomarkers or age. During training each modality is dropped with probability 0.15 (never all at once), so that no single input, particularly age, carries the model.

Two heads. Normal < MCI < Dementia is an ordered scale. Besides a softmax head, an ordinal cumulative-link head [21] predicts a severity score $s_{\text{sev}}$ and two ordered thresholds $t_1 < t_2$, with $P(y > k) = \sigma(s_{\text{sev}} - t_k)$. The class probabilities of the two heads are averaged.

### E. Training

The loss is a class-weighted focal cross-entropy [22] ($\gamma = 1.5$, weights 1.0/1.4/1.2 for Normal/MCI/Dementia, label smoothing 0.05) plus 0.5 times the cumulative binary cross-entropy of the ordinal head. We train for 35 epochs with AdamW (learning rate $10^{-3}$, weight decay 0.05, 3-epoch warm-up then cosine decay), 16 recordings per

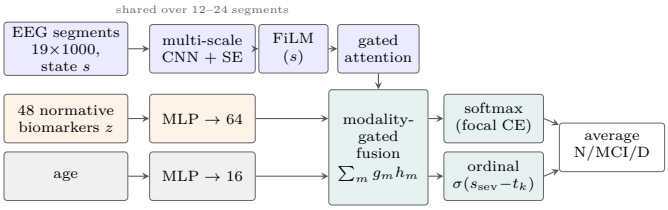



Fig. 2. SANF-Net. State-tagged EEG segments, age-normative biomarkers and age are encoded separately and combined by modality-gated fusion feeding a softmax and an ordinal head.

batch, mixed precision, gradient clipping at 2.0 and an exponential moving average (EMA) of the weights (decay 0.995). Augmentation comprises amplitude scaling (0.9–1.1), Gaussian noise, channel dropout (0.1), biomarker noise and same-class bag mixing: with probability 0.3, 25–50% of a recording's segments are replaced by segments of another training recording with the same label. The network has 143k parameters and trains in about 6 min per fold on a laptop RTX 4050 GPU. We evaluate the final EMA weights rather than selecting a checkpoint on a small validation set.

### F. Evaluation protocol

All metrics are per recording. Our primary protocol is stratified five-fold cross-validation over the pooled 1,069 training and validation recordings, with the normative model and age scaling refitted in each fold, giving one out-of-fold (OOF) prediction per recording. The main models are evaluated with five different fold assignments (repeated CV, $5 \times 5$) and reported as mean ± s.d. over repeats; ablations and subgroup analyses use the first fold assignment. We also report the official test and no-overlap test sets, scored by the mean prediction of the five fold models. Metrics are accuracy, balanced accuracy, macro one-vs-rest AUC, per-class recall and pairwise AUCs, with 95% percentile bootstrap confidence intervals (1,000 resamples of recordings); model differences use a paired bootstrap on the same OOF recordings. Subgroup counts in OOF analyses refer to the 1,069 pooled recordings (229 community normals, 171 SMI).

Baselines, all with exactly the same folds and per-fold normative refitting. Tabular: age alone; the 48 biomarkers, raw or normative; normative biomarkers plus age; each with $L_2$-regularised logistic regression (LR; class-balanced, $C$ tuned by inner five-fold CV) and, for the last, histogram gradient boosting (HGB). Deep, EEG from random crops: EEGNet-8,2 [12] and a CEEDNet-style 1D-ResNet (four stages of two residual blocks on the raw 19 channels, age concatenated before the classifier, 2.3M parameters) [6]. Both use SANF-Net's data pipeline, loss, schedule, augmentation and EMA, and average segment logits per recording; the 1D-ResNet follows CEEDNet's design idea but is not the authors' code or training recipe.

TABLE II
Three-class results (per recording). CV: 5-fold out-of-fold, $n = 1069$ (mean ± s.d. over 5 fold assignments where marked *; 4 for SANF-Net). Test: official test split, $n = 118$, mean of the five fold models of the first assignment.

| | CV out-of-fold | | | Test | | |
|---|---|---|---|---|---|---|
| Model | Acc | BAcc | AUC | Acc | BAcc | AUC |
| Age (LR)* | $51.1 \pm 0.2$ | $51.6 \pm 0.2$ | $.709 \pm .002$ | 55.1 | 54.6 | .739 |
| Biomarkers, raw (LR) | 56.0 | 56.0 | .761 | 63.6 | 63.5 | .815 |
| Biomarkers, normative (LR) | 58.1 | 58.1 | .761 | 62.7 | 63.2 | .809 |
| Norm. biomarkers + age (LR)* | $59.3 \pm 0.7$ | $59.3 \pm 0.7$ | $.778 \pm .004$ | 66.1 | 65.9 | .839 |
| Norm. biomarkers + age (HGB) | 59.2 | 58.9 | .779 | 61.0 | 60.2 | .813 |
| EEGNet [12] | 52.9 | 51.5 | .707 | 60.2 | 58.0 | .773 |
| 1D-ResNet (CEEDNet-style) | 57.7 | 57.1 | .752 | 61.0 | 60.0 | .810 |
| CEEDNet [6]† | – | – | – | 70.2 | – | – |
| SANF-Net (ours)* | $59.5 \pm 0.3$ | $59.1 \pm 0.4$ | $.778 \pm .006$ | 66.1 | 66.1 | .862 |

†As published (own training recipe and test-time augmentation); 74.7% with a 9-model ensemble.

## V. Results

### A. Main results

Table II summarises three-class performance. On the first fold assignment SANF-Net reaches 59.8% OOF accuracy (95% CI 56.7–62.8%) and a macro AUC of 0.780 (0.759–0.801); over four repeated fold assignments it reaches $59.5 \pm 0.3$% and $0.778 \pm 0.006$. On the official test split the five-fold ensemble scores 66.1% (57.6–74.6%) with AUC 0.862, and 64.4% (AUC 0.853) on the no-overlap subset. The test split is easier than the pooled data: the same architecture trained on the official split scores $59.9 \pm 1.0$% on validation but $68.4 \pm 2.8$% on test (3 seeds). Single-split numbers on CAUEEG should therefore be read with intervals of about ±8 points.

The tabular baselines are strong. Age alone gives 51.1% (AUC 0.709), and logistic regression on normative biomarkers plus age reaches $59.3 \pm 0.7$% (AUC $0.778 \pm 0.004$) over five repeats. On the first fold assignment the paired difference from SANF-Net is +0.2 points in accuracy (95% CI −2.7 to +3.3) and +0.003 in AUC (−0.013 to +0.020); over the four shared fold assignments it is $-0.1 \pm 0.5$ points and $-0.001 \pm 0.006$ AUC: the two models are indistinguishable. The deep EEG-only baselines are weaker: EEGNet reaches 52.9% (AUC 0.707) and the 1D-ResNet 57.7% (AUC 0.752). SANF-Net is significantly better than EEGNet (+6.8 points, CI 3.4–10.1) and than the 1D-ResNet in AUC (+0.028, CI 0.003–0.050), though not in accuracy (+2.1, CI −1.4 to +5.4). On the official test split the 1D-ResNet scores 61.0%, below the published 70.2% for CEEDNet, which used a larger model, longer training and test-time augmentation; we do not claim to reproduce it.

MCI is the bottleneck. Recall is 76.8% for Normal, 62.9% for Dementia and only 38.8% (33.9–44.1%) for MCI, whose errors split evenly between Normal and Dementia (Fig. 3). For MCI versus Normal, SANF-Net's AUC is 0.787 (0.752–0.818), detecting 63% of MCI at 80% specificity. Here the linear model is significantly better (AUC 0.823; paired difference −0.037, CI −0.057 to −0.018; Fig. 5, left), and it also has higher MCI recall (47.9% versus 38.8%). Age alone already achieves 0.754.

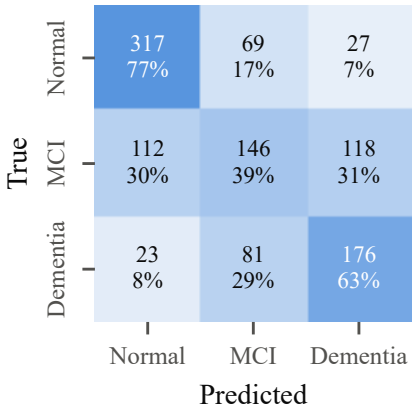



Fig. 3. SANF-Net out-of-fold confusion matrix ($n = 1069$, first fold assignment; counts and row percentages).

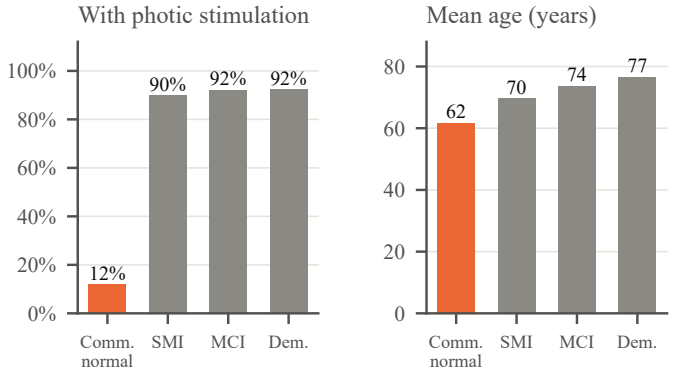



Fig. 4. Share of recordings with photic stimulation and mean age per subgroup. Community normals (orange) differ from all clinical groups in both, independently of cognition.

TABLE III
Ablations (5-fold CV out-of-fold, $n = 1069$, first fold assignment).

| Variant | Acc | AUC | AUC N vs MCI | MCI recall |
|---|---|---|---|---|
| SANF-Net (full) | 59.8 | .780 | .787 | 38.8 |
| random crops, no state | 58.6 | .767 | .781 | 36.2 |
| no FiLM state conditioning | 58.9 | .777 | .784 | 41.8 |
| mean instead of attention pool | 58.5 | .782 | .783 | 35.9 |
| plain standardisation (no normative) | 57.6 | .768 | .768 | 36.4 |
| no biomarkers | 57.3 | .772 | .767 | 34.8 |
| no age | 58.5 | .783 | .788 | 37.0 |
| EEG only | 55.3 | .731 | .719 | 24.5 |
| no ordinal head | 59.5 | .777 | .785 | 58.8 |

Section V-C shows that the linear model's advantage comes from the protocol features.

## B. Ablations

Table III removes one component of SANF-Net at a time under the same protocol (first fold assignment); paired bootstrap CIs are relative to the full model. Only removing both non-EEG branches clearly hurts: the EEG-only network loses 4.5 accuracy points (CI 1.4–8.0), 0.049 macro AUC and 0.068 MCI-versus-Normal AUC, and its MCI recall falls to 24.5%. Every single component is worth at most about two accuracy points, and no single removal is significant in accuracy. State-aware cropping gives a small but significant AUC gain (+0.013, CI 0.002–0.025); normative scoring (+0.018) and the biomarker branch (+0.020) help MCI-versus-Normal AUC with CIs just touching zero. FiLM, attention pooling and the age input are not distinguishable from zero. Removing the ordinal head leaves accuracy and AUC unchanged but moves the operating point: MCI recall rises from 38.8% to 58.8%, at the cost of Normal (76.8 to 68.3%) and Dementia (62.9 to 47.5%) recall. The ordinal loss therefore pulls borderline cases towards the extreme classes rather than adding information.

## C. Protocol and age confounds

Community-recruited normals differ from every other group in ways unrelated to cognition (Table I, Fig. 4). They are younger (61.7 versus 69.6–76.6 years), only 12% of their recordings include photic stimulation (versus 90% of SMI, 92% of MCI and 92% of dementia recordings), fewer include an eyes-open period (63% versus 80–87%), and their recordings are shorter (10.2 versus 13.8–14.0 min). A model can therefore recognise many community normals from the protocol alone: the has_photic and has_eo indicators, the photic-driving feature, and the presence of photic segments in the EEG bag all encode it.

Protocol-blind models. We trained (A) SANF-Net with photic segments removed from the EEG bag and has_photic, has_eo and photic-driving SNR removed from the biomarkers; (B) SANF-Net on eyes-closed segments only, which also removes eyes-open availability from the EEG branch; and the linear model without the same three features. Table IV shows that every model separates community normals from MCI far better than SMI patients from MCI. Removing protocol information costs SANF-Net 2.7 accuracy points (CI +0.1 to +5.1) and 0.016 AUC (0.002–0.030), and costs the linear model 2.5 points and 0.031 AUC (means over five repeats). The loss falls almost entirely on the community-versus-MCI comparison (0.891→0.852 for SANF-Net, 0.951→0.863 for LR), whereas SMI-versus-MCI stays at 0.65–0.66 for every model. Without protocol features the linear model's MCI-versus-Normal advantage disappears: its AUC falls from 0.822 to 0.775, against 0.764 for protocol-blind SANF-Net.

Age-matched analysis. Age and protocol are entangled, because community normals are both younger and differently recorded. We therefore matched each comparison group 1:1 to MCI recordings by age (greedy nearest neighbour without replacement, caliper 2 years) on the OOF predictions and recomputed AUCs (Table V); an age-only model scores 0.50–0.51 on every matched set, confirming the matching. With age removed, the linear model still separates community normals from MCI (0.864), but almost entirely through protocol: without protocol features it drops to 0.615, and protocol-blind SANF-Net (A) to 0.570. On the 24 community normals

TABLE IV
Protocol confound (5-fold CV out-of-fold, $n = 1069$). Comm.: community-recruited normals ($n = 229$); SMI: subjective memory impairment ($n = 171$). LR rows: mean over 5 fold assignments.

| | | | AUC vs MCI | | |
|---|---|---|---|---|---|
| Model | Acc | AUC | Comm. | SMI | SMI recall |
| Age (LR) | 51.1 | .709 | .858 | .618 | 42.7 |
| Norm. biomarkers + age (LR) | 59.3 | .778 | .951 | .655 | 35.7 |
| without protocol features | 56.8 | .747 | .863 | .660 | 50.6 |
| SANF-Net | 59.8 | .780 | .891 | .653 | 57.9 |
| (A) no photic information | 57.1 | .764 | .852 | .649 | 63.7 |
| (B) eyes-closed only | 58.5 | .756 | .842 | .650 | 63.7 |

TABLE V
Age-matched AUC versus MCI (1:1, caliper 2 years) on out-of-fold predictions, with 95% bootstrap CI over matched pairs.

| | Community | Community with photic | SMI |
|---|---|---|---|
| Matched pairs | 127 | 24 | 168 |
| Mean age (group / MCI) | 66.2 / 66.5 | 63.9 / 63.8 | 70.5 / 70.6 |
| Age (LR) | .51 (.50–.52) | .50 (.47–.53) | .50 (.49–.50) |
| Norm. biomarkers + age (LR) | .86 (.82–.91) | .56 (.45–.69) | .57 (.52–.62) |
| without protocol features | .62 (.57–.66) | .59 (.48–.71) | .55 (.51–.59) |
| SANF-Net | .71 (.66–.77) | .50 (.37–.62) | .54 (.50–.58) |
| (A) no photic information | .57 (.53–.61) | .52 (.40–.65) | .54 (.51–.58) |

whose recording did include photic stimulation, matched to MCI with the same protocol, no model exceeds chance (0.50–0.59, all CIs include 0.5). SMI versus age-matched MCI, where protocol is shared, gives 0.54–0.57 for every model.

### D. What the model uses

The mean modality gates differ by true class: for Normal recordings the EEG branch dominates (EEG 0.42, biomarkers 0.29, age 0.29), whereas for MCI and Dementia the biomarker branch dominates (0.40 and 0.42) and EEG falls to 0.28–0.30. Given Table V, the EEG branch's weight on Normal recordings is consistent with it detecting the absence of photic segments rather than normal physiology.

A cautionary note on development. SANF-Net's design was developed on the official split, and the official test set was consulted during development. In particular, we screened additional feature families (aperiodic exponent, permutation entropy and Lempel–Ziv complexity, wPLI and amplitude-envelope connectivity, band-power variability, microstates); adding connectivity improved the official test accuracy (73.7% versus 72.9% for a single model) but gave no gain under five-fold CV (59.3% versus 59.8%), and we did not adopt it. Official-test numbers for SANF-Net are therefore not strictly held out, which is one reason we treat cross-validated OOF results as primary.

### E. External validation

Table VI and Fig. 5 (right) report MCI-versus-normal performance without retraining, using the same five fold models as all CAUEEG results (each paired with its own fold's normative model, predictions averaged). Since

TABLE VI
External MCI vs. normal without retraining (5-fold CV ensemble). AUC with 95% bootstrap CI.

| Cohort | $n$ (MCI) | No calibration | Site calibration |
|---|---|---|---|
| Isfahan 2016 | 27 (11) | .73 (.51–.91) | .76 (.57–.92) |
| Isfahan 2019 | 34 (18) | .64 (.45–.84) | .63 (.44–.81) |
| Isfahan pooled | 61 (29) | .67 (.53–.80) | .70 (.56–.83) |
| GENEEG | 63 (28) | .58 (.44–.72) | .58 (.42–.74) |

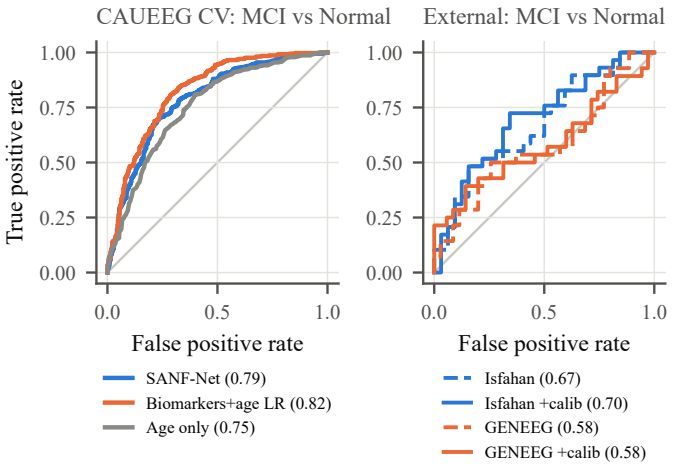



Fig. 5. MCI versus normal ROC curves. Left: CAUEEG five-fold out-of-fold predictions (AUC in brackets). Right: external cohorts without retraining, with and without site calibration.

age is uninformative here, these results reflect EEG and biomarkers only. We also apply a label-free site calibration: channel amplitudes are rescaled to the external cohort's median, and normative $z$-scores are re-centred and re-scaled on the cohort's own normals in a leave-one-out manner, so no subject contributes to its own reference. On the pooled Isfahan cohorts the CV ensemble reaches an AUC of 0.67 (0.53–0.80), rising to 0.70 with site calibration; on GENEEG it is 0.58 (0.44–0.72) with or without calibration, an interval that includes chance. With 61–63 subjects per cohort all intervals are wide. The models trained on the official split (3 seeds) gave similar results (Isfahan pooled 0.649, 0.677 with calibration; GENEEG 0.560, 0.563).

## VI. Discussion

Much of the MCI signal is age and protocol. The headline MCI-versus-Normal AUCs on CAUEEG (0.70–0.82 for every model here) mix three sources: age, recording protocol and EEG physiology. Age alone gives 0.75. After age matching, the clinically relevant comparison, a patient with memory complaints who does or does not have MCI, reaches only 0.54–0.57, and community normals recorded with the same protocol as MCI patients are not distinguished from them. This agrees with Mishra et al.'s observation of strongly overlapping MCI and control EEG feature distributions [11], and suggests a low ceiling

for resting-state spectral EEG as an MCI screen. We recommend that CAUEEG results be reported per Normal subgroup, with age-matched comparisons.

The linear model's MCI advantage is a shortcut. The linear model is significantly better than SANF-Net for MCI versus Normal (0.823 versus 0.787) and recalls more MCI. Tables IV and V show where this comes from: given the protocol indicators, a linear model can identify community normals almost perfectly (age-matched AUC 0.86, 0.90 for those without photic stimulation), which leaves its Normal-versus-MCI boundary free to classify more borderline cases as MCI. Once protocol features are removed, the advantage disappears (0.775 versus 0.764 for protocol-blind SANF-Net). SANF-Net is less dependent on the shortcut, as its higher SMI recall shows (57.9% versus 35.7%).

Architecture versus features. SANF-Net matches but does not beat logistic regression on the same normative biomarkers. It beats the EEG-only deep baselines mainly because it also sees those biomarkers and age: its own EEG-only ablation (55.3%, AUC 0.731) lies between EEGNet and the 1D-ResNet. Gains in the deep-learning literature on this benchmark should therefore be compared against a strong tabular model of established markers and age under the same protocol; without that comparison, gains from architecture cannot be separated from gains from features, age and protocol.

Shortcuts need events to be seen. The protocol confound is invisible to pipelines that take random crops and ignore annotations, yet such models can still exploit it, because photic driving is visible in the raw EEG. Correcting the event clock was a prerequisite both for the state-aware model and for detecting the confound.

Single splits mislead. The official test split is easier than the pooled data by about 8 points, and a feature family that improved the test score showed no gain under cross-validation. Cross-validated OOF estimates with intervals and repeated fold assignments should be the primary CAUEEG metric.

Transfer. Without retraining or age information, the CAUEEG models keep modest MCI discrimination on Isfahan (AUC 0.67–0.70) but little on GENEEG (0.58), whose recordings differ more in montage and protocol. Re-centring normative references on a site's own normals is cheap and label-free, but the external cohorts are too small for firm conclusions.

Limitations. Training data come from a single hospital. Subgroup labels rely on the dataset's symptom codes, and the community-with-photic subgroup is small (24 matched pairs). The external cohorts are small, lack ages and state annotations, and were not recorded with the CAUEEG protocol. Diagnoses are clinical, without biomarker confirmation. The official test set was consulted during model development. Ablations and subgroup analyses use a single fold assignment.

## VII. Conclusion

CAUEEG event annotations need a pause correction before they can be used. Once corrected, they enable state-aware, age-normative modelling of MCI and dementia, and they reveal that community-recruited normals are distinguishable by recording protocol and age alone. SANF-Net, a compact fusion network built on these ideas, reaches 59.5% repeated-CV accuracy and is less dependent on the protocol shortcut than a tabular model, but does not outperform a linear model of normative biomarkers. Once age and protocol are controlled, no model separates SMI or protocol-matched normals from MCI with an AUC above 0.6. We release our event-alignment code and evaluation protocol [TODO: repository URL] to support confound-aware benchmarking of EEG-based cognitive screening.

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